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Characterization of Multidrug-Resistant Biofilm-Producing *Escherichia coli* and *Klebsiella pneumoniae* in Healthy Cattle and Cattle with Diarrhea

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This study describes comparative occurrence and characterization of multidrug-resistant (MDR) *Escherichia coli* and *Klebsiella pneumoniae* (KP) in healthy cattle (HC) and cattle with diarrhea (DC) in India. During 2018–2020, 72 MDR isolates, including 35 *E. coli* (DC: 27; HC 8) and 37 *K. pneumoniae* (DC: 34; HC: 3), from 251 rectal swabs (DC: 219; HC: 32) were investigated for extended-spectrum beta-lactamase (ESBL), AmpC type β-lactamase and carbapenemase production, antimicrobial susceptibility profile, biofilm production, and efflux pump activity. Fifty-five MDR isolates were ESBL producers (ESBLPs) (DC: 50; HC: 5) and ESBLPs from DC were coresistant to multiple antibiotics. The *bla*CTX-M gene (50) was the most frequently detected β-lactamases followed by *bla*AmpC (22), *bla*TEM1 (13), *bla*CMY-6 (6), *bla*OXA1 (5), *bla*PER (2), *bla*DHA, and *bla*FOX and *bla*SHV12 (1 each). Plasmid-mediated quinolone resistance determinants *qnrB*, *qnrS*, *qnrA*, and *qepA* were detected in 18, 16, 2, and 3 isolates, respectively. Twenty three isolates revealed mutation in *gyrA* and *parC* genes. Tetracycline-resistance markers *tetA*, *tetB*, *tetC*, and *tetE* were detected in 33, 10, 3, and 2 isolates, respectively. Only one of the 41 imipenem-resistant isolates harbored *bla*NDM-5 and two were colistin-resistant. Altogether, 20 MDR isolates were strong biofilm producers and 19 harbored different virulence factors. This is the first ever report from India on the presence of MDR Enterobacteriaceae with resistance to even last-resort antimicrobials in the bovine diarrhea.

Keywords: multidrug resistance, ESBL, carbapenem, colistin, bovine

Introduction

HUMAN CIVILIZATION IS at risk of zoonotic infection via direct contact because of proximity with food animals, environmental contamination, and intake of contaminated food items causing enormous economic losses associated with loss of man-days. The situation becomes more complicated due to the transmission of antimicrobial-resistant bacteria such as extended-spectrum beta-lactamase (ESBL)-producing bacteria into the human food chain from the animals. Cattle are considered as a major source of cheap protein as well as an important source of biological fertilizer throughout the world.¹ *Escherichia coli* and *Klebsiella pneumoniae* are enlisted as critical sentinel organisms for antimicrobial resistance in food animals, human and environment due to ubiquitous presence as commensal or

pathogen and horizontal transfer of antibiotic resistance determinants such as ESBLs between the strains.²

ESBLs are essential resistance determinants transmitted through horizontal gene transfer in the Enterobacteriaceae group of commensal or pathogens.³ The production of the enzyme is associated with resistance to a variety of β-lactam antibiotics, including penicillins, higher-generation cephalosporins, and monobactams (*e.g.*, aztreonam), but usually not with carbapenems or cephamycins (*e.g.*, cefoxitin). Moreover, AmpC β-lactamase-producing organisms can produce resistance against cephalosporins, penicillins, cephamycins, and monobactams and cannot be inhibited by β-lactamase inhibitors such as clavulanic acid. The chromosomal *bla*AmpC of *E. coli* is repressed or weakly expressed, although mutation in the promoter region may cause constitutive overexpression. There are three classical ESBLs, that is, TEM (except TEM-1),

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SHV (except SHV-1 and 2), and CTX-M.⁴ Among them, CTX-M is observed as the most prevalent type in clinically infected human patients world wide.⁵

Carbapenem is the last resort antibiotic to treat β -lactamase-producing Enterobacteriaceae.⁶ Carbapenem-resistant *E. coli*/*K. pneumoniae* (CRE) are emerging concern associated with the production of metallo- β -lactamase (VIM), imipenemase (IPM), New Delhi metallo- β -lactamase (NDM-1), oxacillinase (OXA-48), and *K. pneumoniae* carbapenemase (KPC).⁷ On the contrary, colistin (COL) has gained the confidence of the clinicians as a typical drug of choice in the infections caused by CRE. Currently, gram-negative bacteria (GNB) from food animals showing resistance to carbapenem and COL has emerged as a serious concern in several countries, including India.^{8,9} Multidrug-resistant (MDR) *K. pneumoniae* and *E. coli* isolates also possess quinolone resistance genes (PMQR, *qnrA*, *qnrB*, *qnrC*, *qnrD*, and *qnrS*), efflux pumps (*qepA*), sulfonamide resistance (*sul1*, *sul2*, and *sul3*), aminoglycoside-resistance (*ArmA*, *RmtA*, *RmtB*, *RmtC*, *RmtD*, and *NpmA*), and tetracycline-resistance (TC-r) genes (*tetA*, *tetB*, *tetC*, *tetD*, and *tetE*).^{10–14}

Resistance genes are often associated with biofilm formation among clinical isolates of *K. pneumoniae* and *E. coli*.¹⁵ Biofilms are microbial communities adhered to biotic or abiotic surfaces and act as a major site for horizontal gene transfer. The phenotypic transformation from planktonic cells to immobile stage is a coordinated process regulated by environment and putative factors.¹⁶

Food animals such as cattle are considered as reservoirs of MDR bacteria, including ESBL/AmpC-producing *K. pneumoniae* and *E. coli*.¹⁷ Although there is no direct evidence of MDR—Enterobacteriaceae transmission into human food chain from cattle, a few studies elucidated the pseudovertical transmission from parent broiler flocks to hatcheries.¹⁸ This kind of transmission can be further extended into the community through the farm handlers. Generation of MDR bacteria in food animals is associated with direct exposure to antibiotics, which might be more in adult cattle and calves suffering from diarrhea due to frequent therapeutic intervention. Neonatal calf diarrhea (NCD) is a multitierological complex, in which enterotoxigenic or enteropathogenic *E. coli* plays a vital role.¹⁹ The studies associated with occurrence of ESBL/AmpC-producing *Klebsiella* and *E. coli* in cattle are limited,^{20,21} and no study is apparently available to depict a comparative profiling of MDR bacteria from the healthy cattle (HC) and from diarrheic cattle or calves with NCD. Thus, the present study was conducted to detect and characterize MDR *K. pneumoniae* and *E. coli* isolates from rectal swabs of diarrheic and asymptomatic cattle from five agro-climatic zones of West Bengal (India), the state with one of the largest cattle populations in the country with an increasing trend.²²

Materials and Methods

Sample collection, isolation, and identification

Rectal swab samples (n : 219) were aseptically collected from adult cattle (n : 46) and calves with diarrhea (n : 173) from different districts of five agro-climatic zones of West Bengal, India—Vindhyan Alluvial Zone (n : 38), Gangetic Alluvial Zone (n : 67), Undulating Red and Laterite Zone (n : 36), Coastal Saline Zone (n : 51), and Terai-Teesta

Alluvial Zone (n : 29). The animals were maintained in the semi-intensive stall-feeding system with an average herd size of 10–30. The calves suffering with NCD were treated with gentamicin, potentiated amoxicillin, tetracycline, and cephalosporins. Few adult cattle earlier suffered from mastitis and were treated with intramammary infusion of cefoperazone and cefquinome.

Following overnight enrichment, the samples were plated on MacConkey agar, from which 3–5 lactose fermenting colonies were picked up randomly for incubation in nutrient broth to run standard biochemical tests for presumptive identification. Furthermore, the isolates were subjected to molecular tests, a pentaplex PCR covering five genes (*lacY*, *lacZ*, *cydA*, *uidA*, and *phoA*) and a duplex PCR for two genes (*gyrA* and *pehX*) for confirmation of *E. coli*²³ and *K. pneumoniae*,²⁴ respectively. Likewise, rectal swab samples were also collected from healthy or asymptomatic cattle (n : 32) and processed similarly.

Antibiotic susceptibility testing

Confirmed *E. coli* and *K. pneumoniae* isolates were subjected to antimicrobial susceptibility testing (AST) for 11 classes of antimicrobials namely, penicillin (ampicillin, AMP), potentiated penicillin (amoxicillin-clavulanic acid, AMC), aminoglycosides (gentamicin, GEN), cephalosporin [ceftiofur (EFT 30 μ g); cefpodoxime (CPD); ceftazidime (CAZ); cefotaxime (CTX); ceftriaxone (CTR)], monobactam [aztreonam (AZT)], fluoroquinolone (FQ) [nalidixic acid (NAL); enrofloxacin (ENR 30 μ g)], carbapenem (IPM), dihydrofolate reductase inhibitor, and sulfonamide [sulfamethoxazole with trimethoprim (SXT, 23.75/1.25 μ g)], tetracyclines [tetracycline (tet)], phenicols [chloramphenicol (C, 30 μ g)], and polymixin (COL). While disc diffusion test was conducted for AST against ENR, EFT, C, and SXT, broth dilution assay was used for COL-resistance determination. For other antibiotics, agar dilution assay was used. The antibiotic susceptibility of the isolates were determined following the breakpoints prescribed by Clinical and Laboratory Standards Institute (CLSI).²⁵ Isolates resistant to more than three different classes of antimicrobials were categorized as MDR. For each isolate, the multiple antibiotic resistance index (MARI) was calculated as the ratio of the number of antibiotics, to which the isolate exhibited resistance to the number of antibiotics, to which the isolate had been tested.²⁶

ESBL and AmpC enzyme detection

All the MDR isolates, which were resistant to any of the extended-spectrum cephalosporins (ESCs) or monobactam, were evaluated for ESBL and AmpC type β -lactamase production using three phenotypic tests—combination disc (CD) test, cefoxitin-cloxacillin double-disc synergy test (CC-DDS), and Ezy MICTM ESBL-AmpC coexistence detection test (HiMedia, India). In CD test, pairs of discs containing ESCs (cefotaxime or ceftazidime 30 μ g) alone or with CA (30/10 μ g) were placed at a distance of 25 mm, center to center, on a cation-adjusted Mueller Hinton agar (MHA) plate inoculated with the test isolates. An increase in the inhibition zone diameter of 5 mm or more for an ECS+CA disc versus ESC disc alone was indicative of ESBL production.³ Modified Three Dimensional Tests (M3DT) was performed following the way prescribed by

Shahid *et al.*²⁷ with little modification. A cefoxitin disc (30 µg FOX) was placed at the center of an MHA plate inoculated with a lawn of *E. coli* ATCC 25922. Multiple linear slits of 3 cm were made radially, 3 mm away from the edge of FOX disc with a well at end of each slit. The enzyme extract (40 µL) of the test isolates prepared by repeated freeze-thawing was put into the wells. Following 18–24 hr of incubation at 37°C, isolates showing clear distortion at the edge of the slit were noted. To confirm the AmpC production, the test was repeated after incubating the extract with a 5 µg cloxacillin disc for 30 min at 37°C before being discharged in the well. Inhibition of zone distortion following cloxacillin treatment confirmed AmpC production. Ezy MIC ESBL-AmpC coexistence was determined using improved ESBL and AmpC detection Ezy MIC strips following the manufacturer's guideline (HiMedia).

Metallo-β-lactamase detection

To confirm carbapenemase/metallo-β-lactamases (MBL) production, the isolates with reduced IPM susceptibility were further subjected to modified carbapenem inactivation method (mCIM)²⁸ and Carba NP direct tests (CNPt-direct).²⁹ In mCIM assay, a meropenem disc (MEM, 10 µg) was coincubated with test bacteria in 2 mL trypticase soy broth at 35°C for 4 hr and then placed on MHA plate, previously streaked with 0.5 McFarland suspension of *E. coli* ATCC[®] 25922, for further overnight incubation at 35°C. A zone diameter of 6–15 mm or presence of colonies within a 16–18 mm zone (indicative of inactivated MEM) was considered positive for MBL. For CNPt-direct test, 1 µL loop-full bacterial mass, scraped off from its overnight growth, was suspended in two separate 1.5-mL microfuge tubes—(1) reaction tube containing 100 µL of aqueous indicator solution comprising 0.1% (v/v) of Triton X-100, 0.05% phenol red with 0.1 mmol/L ZnSO₄ (pH 7.8) and 12 mg/mL IMP-cilastatin (equivalent to 6 mg of IPM) and (2) control tube having the above-mentioned solution without IPM-cilastatin. The suspension was vigorously mixed and incubated at 35°C for 2 hr for any appreciable color change (red to orange/yellow) in the reaction tube indicating MBL production.

MEM carbonyl cyanide-*m*-chlorophenylhydrazine disc synergy test

All the IMP-R isolates were further incubated on MH agar with two MEM discs placed 20 mm apart—one with 30 µg of MEM and another with 30 µg of MEM and 200 µg CCCP (carbonyl cyanide-*m*-chlorophenylhydrazine) for 24 hr at 37°C. An increase of 5 mm or more in zone diameter around MEM-CCCP disc than that of the MEM disc alone was interpreted as positive for EP-mediated carbapenem-resistance.³⁰

Assessment of efflux pump activity

To investigate the role of the EP in FQ and polymixin resistance, all the ENR/NAL- and COL- nonsusceptible isolates were screened for MIC_{CIP} and MIC_{COL} using the EP inhibitors at fixed subinhibitory concentrations—CCCP and phenylalanine arginine β-naphthylamide (PAβN) (Sigma-Aldrich) as prescribed previously.^{31,32}

Biofilm detection by tissue culture plate method

All the MDR isolates were evaluated for biofilm production using tissue culture plate (TCP) assay,³³ where 100 µL of 100-fold diluted inoculum of 0.5 McFarland units of all the isolates were dispensed in a 96-well TCP containing 100 µL brain–heart infusion broth and incubated for 72 hr at 37°C. After washing the nonadherent cells, the adherent cells were fixed (with 2% sodium acetate) and stained with 0.1% crystal violet for 30 min. Following removal of the excess stain, the dye with the adherent cells was solubilized in ethanol for spectrophotometric measurement of biofilm at 595 nm (Multiskan ELISA Reader, Thermo).

Molecular characterization

All the MDR isolates were screened for the presence of the genes encoding different β-lactamases—*bla*CTX-M, *bla*TEM, *bla*SHV, *bla*VEB, *bla*AmpC, different plasmid-mediated AmpC type β-lactamases (pAmpC), genes conferring carbapenem resistance (CR)- *bla*KPC, *bla*NDM, *bla*OXA-48-like, and *bla*VIM by PCR^{34–36} following extraction of genomic and plasmid DNA. Further, the isolates were screened for the presence of plasmid-mediated quinolone resistance (PMQR) determinants [*qnrA*, *qnrB*, *qnrS*, *qnrC*, *qnrD*, *aac(6c)-Ib-cr*], quinolone EPs (*qepA*, and *oqxA* and *B*), genes mediating tetracycline- (*tetA*, *tetB*, *tetC*, *tetD*, and *tetE*), and sulfonamide resistance (*sul1*, *sul2*, and *sul3*).^{37–40} The quinolone-resistant determinant regions (QRDRs) of *gyrA* and *parC* were also amplified and sequenced.⁴¹ The COL-R isolates (MIC_{COL} ≥ 4 µg/mL) were screened by multiplex PCR for detection of plasmid-based COL resistance genes⁴² and involvement of chromosomal modification if any, as described previously.⁴³ Further, the MDR *E. coli* isolates were screened for the presence of Shiga toxin-producing genes⁴⁴ and extra-intestinal pathogenic factors.⁴⁵ Details regarding oligonucleotides and PCR conditions are described in Table 1. Relative sensitivity and specificity of the PCR used in the study are illustrated in Supplementary Table S1.

Cloning and sequencing

Amplicons were purified with Nucleospin gel and PCR purification kit (Takara Bio, Inc., Japan) and cloning vector pMD20-T (Mighty TA cloning Kit, Takara Bio Inc.) or pTZ57/R (Fermentas) were used to clone some selective amplicons as insert. The plasmids containing the expected insert were sequenced using the Big Dye Terminator_{v3.1} Cycle Sequencing Kit (Applied Biosystems, Inc., Foster City, CA) on an automated sequencer (Applied Biosystems 3130 Genetic Analyzer) following the manufacturer's instructions. After sequencing, homology searches were made using BLAST algorithm available at <http://blast.ncbi.nlm.nih.gov/Blast.cgi>.

The study was conducted following the guidelines prescribed by Institutional Animal Ethical Committee.

Results

All the 219 rectal swab samples from animals with diarrhea yielded characteristic *E. coli* colonies; however, large, shiny, pink, and mucoid colony characteristic for *Klebsiella* were detected only in 142 samples. From each sample, two characteristic colonies for each bacterium (*E. coli* and

TABLE 1. DETAILS OF THE OLIGONUCLEOTIDES AND PCR USED IN THE STUDY

Primer	Oligonucleotide sequence(5'–3')	Amplicon size (bp)	PCR cycle	Reaction details	Internal control ^a	Reference	GenBank
TEM_for	CATTTCGTCGCGCCCTTATTC	800	94 × 10 min/94 × 40 sec—	0.2 mM dNTPs, 1.5 mM	<i>Escherichia coli</i>	36	MT174046
TEM_rev	CGTTCATCCATAGTTCCTGAC		60 × 40 sec—72 × 60 sec	MgCl ₂ , 0.5 μM each	strain—ivkdr6		
			(30 cycles)/72 × 10 min	primer			
SHV_for	AGGATTGACTGCCCTTTTIG	393	94 × 5 min/94 × 30 sec—		<i>Klebsiella pneumoniae</i>	34	MT174044
SHV_rev	ATTGCTGATTTCGCTCG		60 × 30 sec—72 × 30 sec		strain-HOMBK11		
			(20 cycles)/94 × 30 sec—				
			58 × 30 sec—72 × 30 sec				
			(10 cycles)/72 × 10 min				
OXA_rev	GGCACACAGATTCAACTTTCAAG	564	94 × 10 min/94 × 40 sec—		<i>K. pneumoniae</i> 700063	36	MT371583
OXA_rev	GACCCCAAGTTTCTCTGTAAGTG		60 × 40 sec—72 × 60 sec		<i>E. coli</i> strain 15121/b		
			(30 cycles)/72 × 10 min				
CTXMGp1_for	TTAGGAARTGTGCCGCTGYA	688	94 × 10 min/94 × 40 sec—	0.25 mM dNTPs,	<i>K. pneumoniae</i> strain		MT174045
CTXMGp1_rev	CGATACTGTTGGTGTGCCAT		60 × 45 sec—72 × 60 sec	1.5 mM MgCl ₂ ,	JRH4K1, <i>E. coli</i>		MT371584
CTXMGp2_for	CGTTAACGGCACGATGAC	404	(35 cycles)/72 × 10 min	0.3 μM each primer,	strain IVRI ERS DK		
CTXMGp2_rev	CGATACTGTTGGTGTGCCAT			0.2 mM dNTPs,	46 <i>E. coli</i> (Gp1);		
CTXMGp9_for	TCAAGCCTGCCGATCTGGT	561		2 mM MgCl ₂ ,	<i>E. coli</i> strain		
CTXMGp9_rev	TGATTCGCGCGTGAAG			0.25 μM each primer	IVKD91(Gp2); <i>K. pneumoniae</i> strain		
CTX-Mg8/	AACRCRCAGACGCTCTAC	326			GF4K3 (8/25);		
25_for					<i>E. coli</i> strain SL-8C		
CTX-Mg8/	TCGAGCCCGAASGTGYAT				(CTX-M 9)		
25_rev							
ACC_for	CACCTCCAGCGACTTGTAC	346	94 × 10 min/94 × 40 sec—	0.2 mM dNTPs, 2 mM	<i>K. pneumoniae</i> strain	36	NA
ACC_rev	GTTAGCCAGCATCACGATCC		60 × 40 sec—72 × 60 sec	MgCl ₂ , 0.5 μM each	HOMBK11 (MOX);		
FOX_for	CTACAGTGGGGTGGTTT	162	(30 cycles)/72 × 10 min	primer	<i>E. coli</i> strains ivkd6/		MT154086
FOX_rev	CTATTTGGGCCAGGTGA	895			(CMY-6), <i>E. coli</i>		NA
MOX_for	GCAACAACGACAATCCATCCT				strain ivkd74 (CMY-2);		
MOX_rev	GGGATAGGCGTAACCTCCCAA	997			ivkd 105, ivkd		
DHA_for	TGATGGCACAGCAGGATATTC				117 (FOX Gr);		MT174047
DHA_rev	GCTTGTACTCTTTCGGTATTCG	538			ivkd132 (ACC Gr);		MT154088
CIT_for	CGAAGAGGCAATGACCCAGAC	683			ivkd98 (DHA)		NA
CIT_rev	ACGGACAGGGTTAGGATAGT						
EBC_for	CGGTAAGCCGATGTTGCG						
EBC_rev	AGCCTAACCCCTGATACA	399	94 × 10 min/94 × 40 sec—	0.2 mM dNTPs, 2 mM	<i>K. pneumoniae</i> strain	36	NA
GES_for	AGTCGGCTAGACCGGAAAG		60 × 40 sec—72 × 60 sec	MgCl ₂ , 0.5 μM each	BBSCM-1 (VEB),		
GES_rev	TTTGTCCTGCTCAGGAT	520	(30 cycles)/72 × 10 min	primer	<i>E. coli</i> strains		MT154087
PER_for	GTCCTCGATAATGAAAGCGT				ivrikoldenthr57a		
PER_rev	TTCGGCTTGAATCGGCTGA	648			(VEB); ivkd89		NA
VEB_for	CATTCCCGATGCAAGCGT				(GES); ivkd		
VEB_rev	CGAAGTTTCTTTGGACTCTG	399			144(PER)		NA
GES_for	AGTCGGCTAGACCGGAAAG						
GES_rev	TTTGTCCTGCTCAGGAT						

(continued)

TABLE 1. (CONTINUED)

Primer	Oligonucleotide sequence(5'–3')	Amplicon size (bp)	PCR cycle	Reaction details	Internal control ^a	Reference	GenBank
OXA-48_for OXA-48_rev	GCTTGATCGCCCTCGATT GATTTGCTCCGTGGCCGAAA	281	94 × 10 min/94 × 40 sec— 60 × 40 sec—72 × 60 sec (30 cycles)/72 × 10 min	0.3 mM dNTPs, 1.5 mM MgCl ₂ , 0.5 μM each primer	<i>E. coli</i> strain ivkd205	36	NA
IMP_for IMP_rev	TTGACACTCCATTACD GATTGAGAAATTAAGCCACTCT	139		0.5 mM dNTPs, 2 mM MgCl ₂ , 0.5 μM each primer	<i>E. coli</i> strains-VMC/ <i>E. coli</i> /2017/74 (KPC); ivkdr6 (NDM); ivkd139 (IMP); ikd124 (VIM)	36	NA
VIM_for VIM_rev	GATGGTGTGGTCGCATA CGAATGCCAGACCCAG	390					NA
KPC_for KPC_rev	CATTCAAGGGCTTTCTTGCTGC ACGACGGCATAGTCATTGTC	538					NA
blaAmpC_for blaAmpC_rev	CCCCGCTTATAGAGCAACAA TCAAATGGTCGACTTCACACC	631	94 × 5 min/94 × 30 sec— 57 × 30 sec—72 × 30 sec (30 cycles)/72 × 10 min	0.25 mM dNTPs, 2 mM MgCl ₂ , 0.2 μM each primer	<i>K. pneumoniae</i> strain JRFIK2	35	MT174049
qnrA_for qnrA_rev	ATTCTCTACGCCAGGATTG GATCGGCAAGGTTAGGTCA	516	95 × 5 min/95 × 30 sec— 56 × 1 min—72 × 1 min (35 cycles)/72 × 10 min	0.2 mM dNTPs, 2 mM MgCl ₂ , 0.2 μM each primer	<i>E. coli</i> strain JSD/14552/a (<i>qnrA</i>), <i>K. pneumoniae</i> strain-WBK17	38,39	MT371576
qnrB_for qnrB_rev	ACGATGCTGGTAGTTGTC ACGACATTCGTCACTGCAA	469					MT371577
qnrS_for qnrS_rev	TAAATTGGCACCTGTAGGC GCAGTCCAGCAGCGGGTAG	417					MT050524
qepA_for qepA_rev	CTTCTGCCCGAGTATCGTG CGG CGT GGG CTA CCT GAACG GCC GAT CGC GTG AAG TTC CG	218	96 × 5 min/96 × 1 min— 60 × 1 min—72 × 1 min (35 cycles)/72 × 7 min	0.2 mM dNTPs, 2 mM MgCl ₂ , 0.5 μM each primer	<i>K. pneumoniae</i> strain- NAMBK6		MT174048
sul1_for sul1_rev	CGG CGT GGG CTA CCT GAACG GCC GAT CGC GTG AAG TTC CG	436	95 × 5 min/95 × 30 sec— 65 × 1 min—72 × 1 min (35 cycles)/72 × 10 min	0.3 mM dNTPs, 1 mM MgCl ₂ , 1 μM each primer	<i>K. pneumoniae</i> strain— JLF8K1	40,41	MW116081
sul2_for sul2_rev	GCGCTCAAGGCAGATGGCAT GCGTTTGATACGGCACCCGT	293	95 × 5 min/95 × 30 sec— 69 × 1 min—72 × 1 min (35 cycles)/72 × 10 min	0.3 mM dNTPs, 3 mM MgCl ₂ , 1 μM each primer	VMC/ <i>E. coli</i> /2014/47		NA
sul3_for sul3_rev	GAG CAAAGAT TTT TGG AAT CG CTA ACC TAG GGC TTTGGA TAT	750	94 × 5 min/95 × 30 sec— 53 × 30 min—72 × 1 min (30 cycles)/72 × 10 min	VMC/ <i>E. coli</i> /2014/93			NA
gyrA_for gyrA_rev	AAATCTGCCCGTGTCTGTGGT GCCATACCTACGGCGATACC	344	95 × 10 min/95 × 1 min— 55 × 1 min—72 × 1 min (35 cycles)/72 × 10 min	0.2 mM dNTPs, 2 mM MgCl ₂ , 0.5 μM each primer	<i>E. coli</i> strain ATCC 25922	40,41	MT371578
ParC_for ParC_rev	CTGAATGCCAGCGCCAAATT GCGAACGATTTCGGATCGTC	168	95 × 10 min/95 × 1 min— 55 × 1 min—72 × 1 min (35 cycles)/72 × 10 min	0.5 mM dNTPs, 2 mM MgCl ₂ , 1 μM each primer			MT371582
blaNDM-5_for blaNDM-5_rev	ATGGAATTGCCCAATATTATGCAC TCAGCGCAGCTTGTCGGC	813	95 × 5 min/95 × 30 sec— 49 × 30 sec—72 × 30 sec (35 cycles)/72 × 10 min	0.2 mM dNTPs, 1.5 mM MgCl ₂ , 0.5 μM each primer	ivkdr6	79	MT050525
tetA_for tetA_rev	GCT ACA TCC TGC TTG CCT TC CAT AGA TCG CCG TGA AGA GG	210	95 × 5 min/95 × 30 sec— 55 × 60 sec—72 × 60 sec (35 cycles)/72 × 10 min	0.2 mM dNTPs, 2 mM MgCl ₂ , 0.75 μM each primer	<i>K. pneumoniae</i> strain- MUMBK1(<i>tetA</i>); <i>E. coli</i> strains ivrikoldec16b (<i>tetB</i>); ivrikoldrh6a (<i>tetC</i>); 6390/a (<i>tetD</i>); 14372/ GR/b (<i>tetE</i>)	37	MT371586
tetB_for tetB_rev	TTG GTT AGG GGC AAG TTT TG GTA ATG GGC CAA TAA CAC CG	659					MT371580
tetC_for tetC_rev	CTT GAG AGC CTT CAA CCC AG ATG GTC GTC ATC TAC CTG CC	418					MT371585
tetD_for tetD_rev	AAA CCA TTA CGG CAT TCT GC GAC CGG ATA CAC CAT CCA TC	787					NA
tetE_for tetE_rev	AAA CCA CAT CCT CCA TAC GC AAA TAG GCC ACA ACC GTC AG	278					MT371579

(continued)

TABLE 1. (CONTINUED)

Primer	Oligonucleotide sequence(5'-3')	Amplicon size (bp)	PCR cycle	Reaction details	Internal control ^a	Reference	GenBank
Stx 1 for	ATAAATCGCCATTCTGTTGACTAC	180	95 × 5 min/95 × 30 sec—	0.25 mM dNTPs,	<i>E. coli</i> strains JK1 and	44	MW384886
Stx 1 rev	AGAACGCCCCACTGAGATCATC		65 × 1 min—72 × 1 min	1.5 mM MgCl ₂ ,	JK2 (positive for		
Stx 2 for	GGCACTGTCTGAAACTGCTCC	255	(15 cycles)/95 × 30 sec—	0.3 μM each primer	<i>stx1</i> , <i>stx2</i> , <i>eaeA</i> and		
Stx 2 rev	TCGCCAGTTATCTGACATTCTG		60 × 1 min—72 × 1 min		<i>hlyA</i>)		
eaeA for	GACCCGGCACAAAGCATAAGC	384	(20 cycles)/72 × 10 min				
eaeA rev	CCACCTGCAGCAACAAGAGG						
hlyA for	GCATCATCAAGCGTACGTTCC	534					
hlyA rev	AATGAGCCAAAGCTGGTTAAGCT						
lacY for	CTA CCG GTG AAC AGG GTA CG	289	95 × 5 min/95 × 30 sec—	0.2 mM dNTPs, 2 mM	<i>E. coli</i> strain ATCC	23	NA
lacY rev	GTC GCT GAA AAA CGC ACT TC		60 × 30 sec—72 × 1 min	MgCl ₂ , 1.5 U TAQ,	25922 for all the		
lacZ for	ATG AAA GCT GGC TAC AGG	517	(35 cycles)/72 × 10 min	0.5 μM for <i>lacZ</i> , <i>cydA</i> and 1 μM for <i>uidA</i> , <i>phoA</i>	genes		
	AAG G						
lacZ rev	CTC CAC ACA GTT TCG GGT TTT C						
cydA for	CCG TAT CAT GGT GGC GTG TGG	398					
cydA rev	GCC GGC TGA GTA GTC GTG GAA G						
uidA for	CGC CGA TGC AGA TAT TCG	603					
uidA rev	GCT GTG ACG CAC AGT TCA TAG						
phoA for	GGT AAC GTT TCA CCG CAG	468					
	AGT TG						
phoA rev	CAG GGT TGG TAC ACT GTC ATT						
	ACG						
gyrA for	CGC GTA CTA TAC GCCATG AAC	441	95 × 5 min/95 × 30 sec—	0.25 mM dNTPs, 2 mM	<i>Klebsiella oxytoca</i>	24	NA
	GTA		55 × 1 min—72 × 2 min	MgCl ₂ , 0.5 μM each	strain H4T4 (<i>gyrA</i>		
gyrA rev	ACC GTT GAT CAC TTC GGT CAG G		(35 cycles)/72 × 10 min	primer, 0.3 mM	and <i>pehX</i>); <i>K.</i>		
rpoB for	CAA CGG TGT GGT TACTGA CG	108	95 × 3 min/95 × 30 sec—	dNTPs, 2 mM	<i>pneumoniae</i> strains		
rpoB rev	TCT ACG AAG TGG CCG TTT TC		55 × 1 min—72 × 2 min	MgCl ₂ , 0.5 μM each	HOMBK9, ATCC		
pehX for	GAT ACG GAG TAT GCCTTT ACG	343	(35 cycles)/72 × 10 min	primer	700603 (<i>gyrA</i> and <i>rpoB</i>)		
	GTG						
pehX rev	TAG CCT TTA TCA AGC GGA TAC						
	TGG						
papC for	ACGGCTGTACTGCAGGGTGTGGCG	328	94 × 5 min/94 × 1 min—	0.25 mM dNTPs,	<i>E. coli</i> strain ivkd66	45	MW384885
papC rev	ATATCCTTTCTGCAGGGATGCAATA		63 × 1 min—72 × 3 min	2.5 mM MgCl ₂ ,			
			(30 cycles)/72 × 10 min	0.15 μM each primer			
iucD for	AAA ACTGACATCGGATGGC	253	94 × 5 min/94 × 1 min—	0.5 mM dNTPs, 2.5 mM	<i>E. coli</i> strain ivkd73		MW384884
iucD rev	GTATTGTGGCAACGCAGAA		60 × 1 min—72 × 1 min	MgCl ₂ , 1 μM each			
			(35 cycles)/72 × 10 min	primer			
astA for	CCATCAACACAGTATATCCGA	119	95 × 5 min/95 × 1 min—	0.5 mM dNTPs, 2.5 mM	<i>E. coli</i> strain		
astA rev	GGTCGCGAGTGACGGCTTGT		55 × 1 min—72 × 1 min	MgCl ₂ 0.5 μM each	ivrikoldec19a		
			(35 cycles)/72 × 10 min	primer			

^aGenes/groups for which the internal controls used in multiplex reactions are given in parenthesis. Otherwise the internal controls indicate the corresponding primers in the same row. dNTPs, deoxynucleotide triphosphates; NA, not applicable; TAQ, Taq polymerase.

Klebsiella [if positive]) were processed for biochemical confirmation and specifies specific PCR. Finally, one confirmed *E. coli* and one confirmed *K. pneumoniae* (if positive) from each sample—in total, 219 *E. coli* and 111 *K. pneumoniae* isolates—were further included for AST. Likewise, 30 *E. coli* and 19 *K. pneumoniae* were isolated from 32 HC and processed. Sixty-one isolates from bovine diarrhea, that included 27 *E. coli* (caec1–caec27) and 34 *K. pneumoniae* (cakp1–cakp34), were found MDR (Table 2). Of them, forty isolates were coresistant to aminoglycoside (GEN), penicillin (Amp), potentiated penicillin (AMC), cephalosporins (any of the drugs—EFT, CPD, AT, CAZ, CTX, and CTR), tetracycline (Tet), and quinolones (NAL/ENR) and all the forty but seven were not susceptible to

imipenem. For all these isolates, MARI ranged between 0.6 and 1.0. On the contrary, 11 isolates including 8 *E. coli* (caec1–caec8) and 3 *K. pneumoniae* (cahkp1, cahkp2, cahkp3) from asymptomatic animals demonstrated MDR feature with MARI ranging between 0.25 and 0.625. All these 11 isolates were resistant to quinolones (NAL/ENR), 9 to cefpodoxime, 8 to ampicillin, 7 to amoxycylav, 4 to cef-tazidime and 6 to each tetracycline and ceftiofur.

Interestingly, 50 of the 61 MDR isolates (~82%) from bovine diarrhea were ESBL producers (ESBLPs) and these ESBLPs exhibited resistance to fluoroquinolone and tetracycline (100% each), aminoglycosides, and carbapenem (76% each) with MARI ranging from 0.6 to 1.0 (Table 2). Only 12 ESBLPs were strong biofilm producers with MARI

TABLE 2. CHARACTERISTICS OF MULTIDRUG-RESISTANT ISOLATES FROM HEALTHY CATTLE AND CATTLE WITH DIARRHEA

Agro-climatic zone	Multidrug-resistant isolates with resistance characteristics
Vindhyan Alluvial Zone	Diarrhea group caec1 -AmpC- <i>qnrB-tetA</i> (ACBL) ^a ; caekp13 -CTXM1- <i>tetA-sul1</i> (ESBL); caekp14 - <i>qnrS-sul1</i> (ESBL) ^a ; caekp22 -CTXM1- <i>tetA</i> (ESBL); caekp26 -CTXM1- <i>tetA-sul1</i> (ESBL); caec9 -CTXM1 (ESBL); caec10 -AmpC- <i>tetA</i> (ACBL) ^a ; caec14 -CTXM1 (ESBL); caec15 -CTXM1 (ESBL); caekp33 -CTXM1- <i>tetB</i> (ESBL) ^a ; caekp34 -CTXM1- <i>qnrB</i> (ESBL); caec22 -TEM-OXA-CTXM1-CMY-FOX-AmpC- <i>tetA-tetB</i> (ESBL)
Gangetic Alluvial Zone	Healthy group cahec1 -AmpC- <i>qnrB-tetA</i> (ACBL)
	Diarrhea group caec2 -CTXM1- <i>tetA-sul1</i> (ESBL & ACBL); caec4 -CTXM1- <i>tetB-sul1</i> (ESBL & ACBL) ^a ; caec6 -AmpC- <i>tetA-sul1</i> (ACBL) ^a ; caekp1 -CTXM1- <i>qnrS</i> (ESBL); caekp5 -TEM-CTXM1-CMY-AmpC (ESBL); caekp9 -CTXM1- <i>qnrS</i> (ESBL); caekp11 -CTXM1- <i>qnrS</i> (ESBL) ^a ; caekp15 -TEM-CTXM1- <i>qnrS-tetA</i> (ESBL & ACBL); caekp16 -TEM-CTXM1- <i>qnrS-tetA</i> (ESBL); caekp18 -TEM-CTXM1- <i>qnrB</i> (ESBL); caekp20 -CTXM1- <i>qnrB-sul1</i> (ESBL); caekp25 -CTXM1- <i>tetA</i> (ESBL); caec8 -CMY-AmpC (ACBL); caekp27 -CTXM1 (ESBL) ^a ; caekp28 -CTXM1- <i>qnrB-tetA</i> (ESBL); caekp29 -CTXM1- <i>qnrB-AAC(6')-Ib-cr-tetA</i> (ESBL); caec13 -AmpC- <i>tetA</i> (ACBL) ^a ; caekp32 -SHV-CTXM1- <i>tetA</i> (ESBL); caec17 -CTXM1- <i>tetA</i> (ESBL); caec18 -TEM-CMY-NDM- <i>tetA-tetB</i> (MBL); caec20 -OXA-CTXM1-AmpC- <i>tetA-sul1</i> (ESBL & ACBL) ^a ; caec23 -PER-CTXM1- <i>qepA-tetA</i> (ESBL & ACBL); caec24 -PER-CTXM1- <i>qepA-tetB</i> (ESBL & ACBL)
	Healthy group cahec2 -OXA-CTXM1 (ESBL); cahec3 - <i>tetB</i> ; cahec5 -AmpC- <i>tetA</i> ; cahec7 -CTXM1- <i>qnrS</i> (ESBL) ^a
Costal Saline Zone	Diarrhea group caec3 -TEM-AmpC- <i>qnrB-tetB</i> (ACBL) ^a ; caec5 -AmpC- <i>qnrB-tetA</i> (ACBL) ^a ; caec7 -CTXM1- <i>tetA</i> -ESBL (ACBL); caec11 -AmpC- <i>tetA</i> (ACBL) ^a ; caekp2 -TEM-CTXM1- <i>qnrS</i> (ESBL); caekp3 -CTXM1- <i>qnrS</i> (ESBL); caekp4 -TEM-CTXM1- <i>qnrS</i> (ESBL) ^a ; caekp8 -CTXM1- <i>qnrS</i> (ESBL) ^a ; caekp17 -CTXM1-DHA-AmpC- <i>qnrB-sul1</i> (ESBL); caekp23 -CTXM1- <i>qnrS</i> (ESBL); caekp30 -TEM-CTXM1- <i>tetA</i> (ESBL); caekp31 -TEM-CTXM1 (ESBL)
	Healthy group cahkp1 -AmpC- <i>qnrB-tetB</i> (ACBL); cahec4 -OXA-CTXM1-AmpC- <i>qnrB</i> (ESBL & ACBL); cahec6 -CTXM1- <i>tetA</i> (ESBL & ACBL); cahec8 -TEM-CTXM1- <i>qnrS</i> (ESBL) ^a
Terai-Teesta Alluvial Zone	Diarrhea group caekp6 -AAC(6')-Ib-cr (ESBL); caekp19 - <i>qnrB</i> (ESBL); caec19 -CTXM1-AmpC- <i>qepA-tetA-sul1</i> (ESBL & ACBL); caec26 -CTXM9- <i>qnrA-qnrB-tetB-tetE-sul1</i> (ESBL)
	Healthy group cahkp2 -AmpC- <i>tetC</i> ; cahkp3 -AmpC (ACBL)
Undulating Red and Laterite Zone	Diarrhea group caekp7 -TEM-CTXM1- <i>qnrS-tetA</i> (ESBL & ACBL); caekp10 -CTXM1- <i>qnrS</i> (ESBL) ^a ; caekp12 -CTXM1-CMY-AmpC- <i>tetA-tetC</i> (ESBL); caekp21 -CTXM1- <i>qnrB-tetA</i> (ESBL); caekp24 - <i>qnrB</i> (ESBL) ^a ; caec12 -AmpC- <i>tetA</i> (ACBL) ^a ; caec16 -AmpC- <i>qnrS-tetA</i> (ACBL); caec21 -OXA-CTXM1-CMY- <i>qnrA-tetA-tetC-tetE-sul1</i> (ESBL); caec25 - <i>qnrB-sul1</i> (ESBL); caec27 -CTXM1-AmpC- <i>qnrB-tetB</i> (ESBL).

Isolate names and the animal groups are kept bold.

^aIsolates with strong biofilm producers.

ACBL, isolates are AmpC type β -lactamase producers; ESBL, isolates are extended-spectrum β -lactamase producers; MBL, isolates are metallo- β -lactamase producers.

ranging from 0.7 to 1.0. Eleven non-ESBLs (caec1, caec3, caec5, caec6, caec8, caec10, caec11, caec12, caec13, caec16, and caec18) exhibited resistance to tetracycline and nalidixic acid/enrofloxacin in addition to one or more drugs screened in this study with MARI ranging from 0.2 to 0.6 and majority (8) of them were strong biofilm producers. Nineteen of the 61 MDR isolates produced AmpC type β -lactamase. Five (7) of 11 MDR isolates from asymptomatic animals were ESBLs and were mostly resistant to amoxycylav (~64%), ampicillin (~73%), cefpodoxime (~82%), tetracycline and ceftiofur (54%), and ceftazidime (~37%). Eight of these isolates were AmpC type β -lactamase producer. None of these isolates was resistant to IPM, COL and chloramphenicol.

Of all the 72 MDR isolates investigated for the role of EP in quinolone resistance, only 9 isolates—2 *E. coli* and 7 *K. pneumoniae* showed a characteristic reduction in MIC_{CIP} in the presence of EP inhibitor.

The blaCTX-M gene (45) (MT174045) (CTX-M 1 in 44 and CTX-M 9 (MT371584) in 1 isolate) was more frequently detected among the isolates from bovine diarrhea followed by blaTEM 1 (12) (MT174046), blaOXA-1 (3) (MT371583), blaPER (2) (MT154087), and blaSHV12 (1) (MT174044) genes. Other β -lactamase genes, blaAmpC (MT174049) was detected in 16 isolates, blaCMY-6 (MT154085, MT154088) in 6 isolates, and blaDHA (MT174047) and blaFOX (MT154086) in one isolate each. PMQR determinants, qnrS (MT050524), qnrB (MT371577), and qnrA (MT371576), were detected in 14, 15, and 2 isolates, respectively. Three isolates carried qepA gene (MT174048). Of all the ENR/NAL-resistant *K. pneumoniae* (12) and *E. coli* (14) isolates, wherein no PMQR gene could be detected, 10 *K. pneumoniae* and 11 isolates *E. coli* revealed mutations in gyrA (MT371578) and ParC gene (MT371582) indicating the role of QRDR-mutation in ENR/NAL-resistance. No other β -lactamase gene except blaCTX-M 1 (5) and blaAmpC (6) was detected in 9 MDR isolates from healthy animals; however, few carried qnrB (3) and qnrS (2) genes. Two isolates from healthy had mutations in gyrA and ParC gene.

All the MDR isolates from diarrheic animals were tet-resistant with Tc-r markers—tetA (MT371586), tetB (MT371580), tetC (MT371585), and tetE (MT371579) in 33, 10, 3, and 2 isolates, respectively. In contrast, only four of the MDR isolates from the healthy group were tet-resistant and the genes tetA and tetB were detected in three and two isolates, respectively. Of 20 sul-resistant isolates from bovine diarrhea, 13 harbored sul1 gene (MW116081). None of the MDR isolates from the healthy group had sul1 gene.

Although 41 isolates were IPM-resistant/nonsusceptible, only one of them (caec18) harbored blaNDM-5 gene (MT050525) and exhibited carbapenemase production in carba NP and mCIM tests. Apart from β -lactams, the isolate was resistant to fluoroquinolone, tet, and aminoglycoside. However, it was susceptible to COL and chloramphenicol. Eleven of these IPM-R isolates have MARI ranging from 0.7 to 1.0 with coresistance to NAL, ENR, TET, and β -lactams. Ten of the IPM-R isolates were strong biofilm producers. Efflux pump-mediated CR was noticed in three isolates when IPM-R isolates were tested against MEM in the presence of CCCP. Further, many of these IPM-R isolates were ESBL (38) or AmpC type β -lactamase producers (8).

In total, two isolates from bovine diarrhea, one *K. pneumoniae* (cakp34) and one *E. coli* (caec21), were

COL-resistant and five were chloramphenicol-resistant (caec8, cakp34, caec21, caec25, and caec26). Upon PCR-screening, none of the COL-resistant isolates had mcr gene(s). However, PCR could not amplify mgrB gene in cakp34. We could not detect any change in MIC_{COL} when the two COL-R isolates were tested in the presence of EPI. Of various virulence factors, estA, iucD, and papC were detected among 11, 4, and 3 isolates from bovine diarrhea (MW384885, MW384884). Only two isolates from this group harbored Shiga toxin-producing *E. coli* virulence genes stx2 (MW384886); however, none of the isolates from healthy animals had any virulence factors.

Discussion

This study attempts to detect and characterize MDR isolates from rectal swab of the adult cattle and calves from the five agro-climatic zones of West Bengal, India. In total, 61 MDR isolates were recovered from 219 rectal swab samples of diarrheic animals (61/219, 27.8%), which included both *E. coli* ($n=27$) and *K. pneumoniae* ($n=34$). These isolates were resistant or nonsusceptible to three or more antimicrobial classes—aminoglycosides (GEN), penicillin (AMP), potentiated penicillin (AMC), cephalosporins (EFT, CPD, AT, CAZ, CTX, and CTR), tetracycline (TET), fluoroquinolones (NAL and ENR), and carbapenem (IPM). Previous studies conducted at different parts of the world noted such MDR *E. coli* strains (to ampicillin, tetracycline, streptomycin, and sulfonamide) in healthy or diarrheic cattle.^{20,46–48} Of late, there was a report of imipenem-resistant *E. coli* in slurry samples from dairy farms²¹; yet, CR in bovine is largely considered incidental. We observed some difference in antimicrobial susceptibility profile of the MDR isolates from asymptomatic animals than that from the animals with diarrhea. Interestingly, while all the MDR isolates from diarrheic animals were tetracycline resistant, only 6 of the 13 isolates from healthy animals exhibited TC-r. Similarly, gentamicin resistance rate of the MDR isolates from diarrheic cattle stood at 73%, much higher than that (36.3%) of the isolates from healthy animals. In contrast to 27.8% SXT resistance of the MDR isolates from diarrheic animals, none of the isolates from healthy animals exhibited SXT resistance. We were unable to find any such comparative study to support our finding.

About 22.8% of the diarrheic animals (50/219) carried ESBLs in contrast to 15.6% of the HC (5/32); this is quite higher than what we observed in cow and buffalo milk in our previous studies.^{24,49,50} The occurrence of ESBL-producing *E. coli* in healthy or diarrheic cattle varies widely (3–63%) in different geographical locations with diverse antibiotic exposure, husbandry practices, and isolation techniques.^{51–53}

Most of the ESBL (+ve) isolates from bovine diarrhea cases exhibited resistance to nalidixic acid/enrofloxacin, tetracycline (100% each), gentamicin (76%), and imipenem (80%) with MARI ranging from 0.6 to 1.0. All five ESBLs from healthy animals were nalidixic acid/enrofloxacin resistant; however, two of them were not susceptible to gentamicin and tetracycline. ESBL-producing *E. coli* strains from bovine displayed similar resistance feature indicating the involvement of multiresistance plasmid in spreading extensive drug resistance.^{21,54,55} Although we had a very small size of ESBL (+ve) isolates from healthy animals, resistance feature of the ESBLs from two (healthy and

diarrheic) population was conspicuously different. Many of the farm owners usually treat the diarrheic animals with antimicrobials, antiparasitic drugs, or home-made ethnoveterinary practices. Prior antimicrobial application of the diarrheic animals may be a possible reason. Nonetheless, such a conclusion requires complete and authentic information on antimicrobial exposure. Most of these samplings were conducted in unorganized sector, and we could not trace the link as many of the animal owners failed to provide meticulous record.

Among various drug-resistant genes, *bla*CTX-M 1 was the predominantly detected β -lactamase gene followed by *bla*-TEM1, *bla*OXA1, *bla*PER, *bla*SHV12, *bla*AmpC, *bla*CMY-6, and *bla*DHA in bovine diarrhea. In healthy animals, *bla*CTX-M 1 *bla*OXA, and *bla*AmpC genes were detected. Cattle are considered as reservoir of *E. coli* strains possessing CTX-M 1, -14, and -15 in Europe^{56,57} and Asia.⁵¹ Even though previous studies from Europe and Africa in bovine rarely reported *bla*SHV gene,^{21,52} it is not an uncommon observation in Asia.⁵⁸ Similar to our findings, chromosomal AmpC and *bla*OXA were detected in bovine *E. coli* strains.²¹ A recent study conducted with human *K. pneumoniae* and *E. coli* strains from four tertiary care hospitals of India indicated the predominance of *bla*TEM, *bla*CTX-M 15, and *bla*OXA-1 genes.⁵⁹ The gene CTX-M 9 detected in one isolate from bovine diarrhea was reported in neonatal stool samples by a group of workers from Kolkata, West Bengal.⁶⁰ It is noteworthy that we were unable to trace any of the β -lactamase genes in five MDR isolates—four *K. pneumoniae* (cakp6, cakp14, cakp19, and cakp24) and one *E. coli* (caec25), which produced ESBL in phenotypic assays. In our previous study too, we noted similar findings among a few ESBL-producing *K. pneumoniae* strains from bovine milk.²⁴ Such findings emphasized on the need for further investigation with such isolates circulating in food animals in this region to explore the hidden mechanism of β -lactam resistance.

Remarkably, except one isolate, we could not detect metallo- β -lactamase/carbapenemase in any of the 43 imipenem nonsusceptible isolates. Only in three isolates, we observed CCCP-mediated potentiation of MEM-inhibition zone indicating the role of efflux-pumps like AcrA in CR.⁶¹ Further, strong biofilm production observed in 10 imipenem-resistant isolates might have a role in CR.⁶² Previous studies indicated that chromosomal and plasmid-mediated cephalosporinase combined with decreased drug permeability might lead to imipenem resistance among the noncarbapenemase producers.⁶³ Of late, Webb *et al.*⁶⁴ reported the role of *bla*CMY-2 in CR among CRE from dairy cattle in the United States. In this study also, we detected pAmpC- β -lactamase genes like CMY-2, FOX, and CMY-6 in some of the imipenem-resistant isolates. Although the *bla*NDM-5-producing GNB were reported in human patients from tertiary care hospitals in the Indian subcontinent, such report in food animals is scarce; Ghatak *et al.*, 2013 reported such pathogen in bovine mastitis.⁹ Further, *bla*NDM-5-positive *K. pneumoniae* were reported from dairy cows in Jiangsu, China.⁶⁵ Carbapenems are not used in food animals in India, however, emergence of CR is a serious concern to public health. The present NDM-5-producing strain is also coresistant to other drugs such as fluoroquinolones, tetracycline, and aminoglycosides harboring the corresponding resistance genes and use of any of these antibiotics could

have facilitated the coacquisition of these resistance genes together with *bla*NDM-5 due to possible genetic linkage. On the contrary, overuse of carbapenem in human could have driven the spread of such pathogens in animals via environmental contamination.⁶⁶

Recently, plasmid-mediated COL resistance mechanism emerged among various GNB pathogens in food animals. However, we were unable to detect such a resistance mechanism in any of the two COL-R isolates in this study. In the COL-resistant *K. pneumoniae*—cakp34, *mgrB* could not be amplified with two different sets of primers indicating the absence of *mgrB* gene. Deletion or inactivation of the *mgrB* gene was described responsible for acquired COL resistance in *Klebsiella*, as it upregulates PhoQ-PhoP and activates the Pmr system responsible for modification of the lipopolysaccharide.^{43,67} However, we could not retrieve such a report in *Klebsiella* sp of animal origin.

None of the cefotaxime-resistant CTX-R (56) or ceftazidime-resistant (55) isolates from this study was sensitive to quinolone. In total, ~72.2% of the isolates were coresistant to cefotaxime/ceftazidime and enrofloxacin/nalidixic acid. Further, 28 of the 50 (56%) *bla*CTX-M harboring ESBLs carried any of the PMQR genes investigated in this study. This finding reiterated the reports of the previous workers describing a strong correlation between ESBL production and quinolone resistance in human⁶⁸ and animals²⁴ possibly due to their carriage on the same plasmid.

Abundance of PMQR genes documented in this study is in corroboration with our previous observations with the ESBL-producing Enterobacteriaceae from bovine and bubaline milk.^{24,49,50} The drug efflux pump-*qepA* was detected in three isolates; all of which exhibited considerably higher MIC to nalidixic acid (results not shown). The gene *qepA* is an antiporter quinolone EP system having considerable similarity to the MFS-type efflux pumps, which actively excrete the drugs from the cytosol to the exterior of bacterial cells.⁶⁹ On the other hand, the gene *AAC(6')-Ib-cr*, detected in two of the quinolone-resistant isolates, was reported to encode the aminoglycoside acetyltransferase that conferred reduced susceptibility to fluoroquinolone by *N*-acetylation of its piperazinyl amine. Mutation in the QRDR (*gyrA* and *ParC*) that was detected in 21 MDR isolates from bovine diarrhea (34.4%, 21/61) and 4 isolates from asymptomatic animals had been reported to cause fluoroquinolone-resistance.⁷⁰ Only 12.8% of the MDR isolates exhibited efflux pump mediated fluoroquinolone-resistance when tested for MIC_{CIP} in the presence of EPI (CCCP); this could have additionally contributed to FQ-R.³¹ In general, efflux pump-mediated fluoroquinolone resistance is more common among the nonfermenter than Enterobacteriaceae.⁷¹ Moreover, expression of EP might not have been evident in some of the fluoroquinolone-resistant isolates where other resistance mechanisms (like a mutation in QRDR) were involved.⁷²

All but five isolates in this study were tetracycline-resistant (93%, 67/72) and *tetA* gene (33) was predominantly detected followed by *tetB* (10), *tetC* (3) *tetD*, and *tetE* (two each). Such a high frequency of TC-r markers was previously reported among the enterohemorrhagic *E. coli* isolates from bovine in United States.¹³ Das *et al.*⁷³ reported GNB harboring *tetA* and *tetB* genes in the milk of the cows with subclinical mastitis from India. Further, a study conducted in Sweden revealed that tetracycline-resistant *E. coli* were

more frequently associated with diarrheic calves, possibly due to coselection of TC-r in *E. coli* in calves for some “beneficial mutation.”²⁰ Moreover, because of broad-spectrum activity, tetracycline is often used in veterinary practice and that may be an important driver for selection of tetracycline-resistant strains in bovine.⁷⁴ A study conducted in dairy farms of Czech Republic revealed the abundance of TC-r genes in the in fresh excrements of calves and soils in farm proximity and such abundance was linked to the prophylactic use of chlortetracycline in the farms.⁷⁵ In contrast to our observation, Wilkerson *et al.*¹³ reported predominance of *tetB* gene among *E. coli* isolates from cattle with no trace of *tetD* and *tetE* genes. The predominance of *tetA* and *tetB* genes among commensal *E. coli* isolates from cattle farms was also reported in recent past.⁷⁶ Lack of tet genes in many of the tet-resistant strains in the present study indicated that they might harbor other unexplored resistance mechanisms as reported recently by a group of workers from Tunisia.⁷⁷

Apart from COL (all but 2 were sensitive), the isolates were relatively sensitive to chloramphenicol (83.3%, 60/72) and sulfamethoxazole and trimethoprim (76.38%, 55/72). Chloramphenicol was once the drug of choice for treating typhoid and other enteric fever; however, its use has been discontinued after the introduction of safer drugs. Ban of chloramphenicol in food animals, coupled with its restricted use in human beings probably facilitated the bacterial population to become sensitive to it. This finding was also observed in our previous studies.^{24,78}

Among various virulence factors, none of the MDR *E. coli* isolates except two carried *stx*, *eaeA*, and *exhA* genes. Two isolates, caec3 and caec16, were detected with *stx2* and *stx2-eaeA* genes. Although these two isolates were non-ESBL producing, they had PMQR (*qnrB* and *qnrS*) and TC-r (*tetA* and *tetB*) genes. Previously we could not find Shiga toxin-producing gene among the isolates harboring ESBL, PMQR, or Tet-r determinants. Nevertheless, various ExPEC-specific virulence factors were detected among many of the isolates from diarrheic animals as reported previously.⁵⁰ Existence of virulent and resistant determinants in the same isolate in food animals may cause serious food-borne infection in human beings in absence of adequate precaution.

The present study described a comparative higher occurrence of MDR *E. coli* and *K. pneumoniae* in bovine diarrhea than healthy group reflecting the antibiotics usage pattern in livestock. The frequency of resistance determinants, including ESBLs (cattle with diarrhea [DC]: 82% vs. HC: 45%), was found to be higher among the MDR isolates from diarrheic cattle than that from the healthy counterparts. The β -lactamase (CTX-M, AmpC, TEM) was the most common mechanism of resistance; however, coresistance to tetracycline, quinolone, and imipenem was also evident. Isolation of carbapenem- and COL-resistant isolates is worrisome as these are the last resort antibiotics in human health care.

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Supplementary Material

Supplementary Table S1

References

- Alexandratos, N., and J. Bruinsma. 2012. Global perspective studies team. World Agric. Towards 2030:2050.
- Tacconelli, E., and N. Magrini. 2020. Global priority list of antibiotic-resistant bacteria to guide research, discovery, and development of new antibiotics. Available at: https://www.who.int/medicines/publications/WHO-PPL-Short_Summary_25Feb-ET_NM_WHO.pdf (Accessed March 1, 2021).
- Andrews, J. 2012. Detection of Extended-Spectrum β -Lactamases (ESBLs) in *E. coli* and *Klebsiella* Species. British Society for Antimicrobial Chemotherapy. Available at: <http://bsac.org.uk/wp-content/uploads/2012/02/Ecoliklebsiella.pdf> (Accessed March 1, 2021).
- Carattoli, A., T. Coque, H. Hasman, *et al.* 2011. Scientific opinion on the public health risks of bacterial strains producing extended-spectrum β -lactamases and/or AmpC β -lactamases in food and food-producing animals 1 EFSA Panel on Biological Hazards (BIOHAZ). EFSA J. 9:2322.
- Carattoli, A. 2013. Plasmids and the spread of resistance. Int. J. Med. Microbiol. 303:298–304.
- Bradley, J.S., J. Garau, H. Lode, K.V. Rolston, S.E. Wilson, and J.P. Quinn. 1999. Carbapenems in clinical practice: a guide to their use in serious infection. Int. J. Antimicrob. Agents 11:93–100.
- Markogiannakis, A., L.S. Tzouveleakis, M. Psychogiou, E. Petinaki, and G.L. Daikos. 2013. Confronting carbapenemase-producing *Klebsiella pneumoniae*. Future Microbiol. 8: 1147–1161.
- Ghafur, A., C. Shankar, P. GnanaSoundari, *et al.* 2019. Detection of chromosomal and plasmid-mediated mechanisms of colistin resistance in *Escherichia coli* and *Klebsiella pneumoniae* from Indian food samples. J. Glob. Antimicrob. Resist. 16:48–52.
- Ghatak, S., A. Singha, A. Sen, *et al.* 2013. Detection of new delhi metallo-beta-lactamase and extended-spectrum beta-lactamase genes in *Escherichia coli* isolated from mastitic milk samples. Transbound. Emerg. Dis. 60:385–389.
- Chopra, I., and M. Roberts. 2001. Tetracycline antibiotics: mode of action, applications, molecular biology, and epidemiology of bacterial resistance. Microbiol. Mol. Biol. Rev. 65:232–260.
- Hammerum, A.M., D. Sandvang, S.R. Andersen, *et al.* 2006. Detection of sul1, sul2 and sul3 in sulphonamide resistant *Escherichia coli* isolates obtained from healthy humans, pork and pigs in Denmark. Int. J. Food Microbiol. 106:235–237.
- Wachino, J., K. Shibayama, H. Kurokawa, *et al.* 2007. Novel plasmid-mediated 16S rRNA m1A1408 methyltransferase, NpmA, found in a clinically isolated *Escherichia coli* strain resistant to structurally diverse aminoglycosides. Antimicrob. Agents Chemother. 51:4401–4409.

13. Wilkerson, C., M. Samadpour, N. van Kirk, and M.C. Roberts. 2004. Antibiotic resistance and distribution of tetracycline resistance genes in *Escherichia coli* O157:H7 isolates from humans and bovines. *Antimicrob. Agents Chemother.* 48:1066–1067.
14. Yue, L., H.-X. Jiang, X.-P. Liao, *et al.* 2008. Prevalence of plasmid-mediated quinolone resistance qnr genes in poultry and swine clinical isolates of *Escherichia coli*. *Vet. Microbiol.* 132:414–420.
15. Alcántar-Curiel, M.D., C.A. Ledezma-Escalante, M.D. Jarillo-Quijada, *et al.* 2018. Association of antibiotic resistance, cell adherence, and biofilm production with the endemicity of nosocomial *Klebsiella pneumoniae*. *Biomed. Res. Int.* 2018:7012958.
16. Hall, C.W., and T.-F. Mah. 2017. Molecular mechanisms of biofilm-based antibiotic resistance and tolerance in pathogenic bacteria. *FEMS Microbiol. Rev.* 41: 276–301.
17. Hiroi, M., F. Yamazaki, T. Harada, *et al.* 2012. Prevalence of extended-spectrum β -lactamase-producing *Escherichia coli* and *Klebsiella pneumoniae* in food-producing animals. *J. Vet. Med. Sci.* 74:189–195.
18. Daehre, K., M. Projahn, T. Semmler, U. Roesler, and A. Friese. 2018. Extended-spectrum beta-lactamase-/AmpC beta-lactamase-producing Enterobacteriaceae in broiler farms: transmission dynamics at farm level. *Microb. Drug Resist.* 24:511–518.
19. Gulliksen, S.M., E. Jor, K.I. Lie, *et al.* 2009. Enteropathogens and risk factors for diarrhea in Norwegian dairy calves. *J. Dairy Sci.* 92:5057–5066.
20. de Verdier, K., A. Nyman, C. Greko, and B. Bengtsson. 2012. Antimicrobial resistance and virulence factors in *Escherichia coli* from Swedish dairy calves. *Acta Vet. Scand.* 54:2.
21. Ibrahim, D.R., C.E.R. Dodd, D.J. Stekel, S.J. Ramsden, and J.L. Hobman. 2016. Multidrug resistant, extended spectrum β -lactamase (ESBL)-producing *Escherichia coli* isolated from a dairy farm. *FEMS Microbiol. Ecol.* 92: fiv013.
22. Govt of India. 2019. Provisional Key Results of 20th Livestock Census | Department of Animal Husbandry & Dairying. Available at <http://dahd.nic.in/division/provisional-key-results-20th-livestock-census> (Accessed March 1, 2021).
23. Modak, R., S. Das Mitra, P. Krishnamoorthy, *et al.* 2012. Histone H3K14 and H4K8 hyperacetylation is associated with *Escherichia coli*-induced mastitis in mice. *Epigenetics* 7:492–501.
24. Koovapra, S., S. Bandyopadhyay, G. Das, *et al.* 2016. Molecular signature of extended spectrum β -lactamase producing *Klebsiella pneumoniae* isolated from bovine milk in eastern and north-eastern India. *Infect. Genet. Evol.* 44:395–402.
25. CLSI. 2018. Performance standards for antimicrobial disk susceptibility tests; approved standard. Clinical and Laboratory Standards Institute, Wyane, PA.
26. Davis, R., and P.D. Brown. 2016. Multiple antibiotic resistance index, fitness and virulence potential in respiratory *Pseudomonas aeruginosa* from Jamaica. *J. Med. Microbiol.* 65:261–271.
27. Shahid, M., A. Malik, M. Agrawal, and S. Singhal. 2004. Phenotypic detection of extended-spectrum and AmpC β -lactamases by a new spot-inoculation method and modified three-dimensional extract test: comparison with the conventional three-dimensional extract test. *J. Antimicrob. Chemother.* 54:684–687.
28. Pierce, V.M., P.J. Simner, D.R. Lonsway, *et al.* 2017. Modified carbapenem inactivation method for phenotypic detection of carbapenemase production among Enterobacteriaceae. *J. Clin. Microbiol.* 55:2321–2333.
29. Laolerd, W., Y. Akeda, L. Preeyanon, P. Ratthawongjirakul, and P. Santanirand. 2018. Carbapenemase-producing carbapenem-resistant Enterobacteriaceae from Bangkok, Thailand, and their detection by the Carba NP and modified carbapenem inactivation method tests. *Microb. Drug Resist.* 24:1006–1011.
30. Osei Sekyere, J., and D.G. Amoako. 2017. Carbonyl cyanide m-chlorophenylhydrazine (CCCP) reverses resistance to colistin, but not to carbapenems and tigecycline in multidrug-resistant Enterobacteriaceae. *Front. Microbiol.* 8: 228.
31. Hamed, S.M., W.F. Elkhatab, H.A. El-Mahallawy, M.M. Helmy, M.S. Ashour, and K.M.A. Aboshanab. 2018. Multiple mechanisms contributing to ciprofloxacin resistance among Gram negative bacteria causing infections to cancer patients. *Sci. Rep.* 8:12268.
32. Hasdemir, U.O., J. Chevalier, P. Nordmann, J.-M. Pagès, J.-M. Pages, and J.-M. Pagès. 2004. Detection and prevalence of active drug efflux mechanism in various multidrug-resistant *Klebsiella pneumoniae* strains from Turkey. *J. Clin. Microbiol.* 42:2701–2706.
33. Panda, P.S., U. Chaudhary, and S.K. Dube. 2016. Comparison of four different methods for detection of biofilm formation by uropathogens. *Indian J. Pathol. Microbiol.* 59: 177–179.
34. Ogbolu, D.O., O.A. Terry Alli, M.A. Webber, A.S. Oluremi, and O.M. Oloyede. 2018. CTX-M-15 is established in most multidrug-resistant uropathogenic Enterobacteriaceae and Pseudomonaceae from hospitals in Nigeria. *Eur. J. Microbiol. Immunol.* 8:20–24.
35. Shahid, M. 2010. Citrobacter spp. Simultaneously harboring blaCTX-M, blaTEM, blaSHV, blaampC, and insertion sequences IS26 and orf513: an evolutionary phenomenon of recent concern for antibiotic resistance. *J. Clin. Microbiol.* 48:1833–1838.
36. Dallenne, C., A. Da Costa, D. Decré, C. Favier, and G. Arlet. 2010. Development of a set of multiplex PCR assays for the detection of genes encoding important β -lactamases in Enterobacteriaceae. *J. Antimicrob. Chemother.* 65:490–495.
37. Ng, L.-K.K., I. Martin, M. Alfa, and M. Mulvey. 2001. Multiplex PCR for the detection of tetracycline resistant genes. *Mol. Cell. Probes* 15:209–215.
38. Robicsek, A., J. Strahilevitz, D.F. Sahm, G.A. Jacoby, and D.C. Hooper. 2006. qnr prevalence in ceftazidime-resistant Enterobacteriaceae isolates from the United States. *Antimicrob. Agents Chemother.* 50:2872–2874.
39. Qin, T., H. Qian, W. Fan, *et al.* 2017. Newest data on fluoroquinolone resistance mechanism of *Shigella flexneri* isolates in Jiangsu Province of China. *Antimicrob. Resist. Infect. Control* 6:97.
40. Karczmarczyk, M., Y. Abbott, C. Walsh, N. Leonard, and S. Fanning. 2011. Characterization of multidrug-resistant *Escherichia coli* isolates from animals presenting at a University Veterinary Hospital. *Appl. Environ. Microbiol.* 77:7104–7112.
41. Rodríguez-Martínez, J.M., C. Velasco, A. Pascual, I. García, and L. Martínez-Martínez. 2006. Correlation of

- quinolone resistance levels and differences in basal and quinolone-induced expression from three *qnrA*-containing plasmids. Clin. Microbiol. Infect. 12:440–445.
42. Rebelo, A.R., V. Bortolaia, J.S. Kjeldgaard, *et al.* 2018. Multiplex PCR for detection of plasmid-mediated colistin resistance determinants, *mcr-1*, *mcr-2*, *mcr-3*, *mcr-4* and *mcr-5* for surveillance purposes. Euro Surveill. 23:17-00672.
 43. Cannatelli, A., T. Giani, M.M. D'Andrea, *et al.* 2014. MgrB inactivation is a common mechanism of colistin resistance in KPC-producing *Klebsiella pneumoniae* of clinical origin. Antimicrob. Agents Chemother. 58:5696–5703.
 44. Paton, A.W., and J.C. Paton. 2002. Direct detection and characterization of shiga toxigenic *Escherichia coli* by multiplex PCR for *stx1*, *stx2*, *eae*, *ehxA*, and *saa*. J. Clin. Microbiol. 40:271–274.
 45. Ewers, C., T. Janßen, S. Kießling, H.-C. Philipp, and L.H. Wieler. 2005. Rapid detection of virulence-associated genes in avian pathogenic *Escherichia coli* by multiplex polymerase chain reaction. Avian Dis. 49:269–273.
 46. Berge, A.C.B., E.R. Atwill, and W.M. Sischo. 2005. Animal and farm influences on the dynamics of antibiotic resistance in faecal *Escherichia coli* in young dairy calves. Prev. Vet. Med. 69:25–38.
 47. Dolejšká, M., D. Senk, A. Cízek, J. Rybářková, O. Sychra, and I. Literák. 2008. Antimicrobial resistant *Escherichia coli* isolates in cattle and house sparrows on two Czech dairy farms. Res. Vet. Sci. 85:491–494.
 48. Sato, K., P.C. Bartlett, and M.A. Saeed. 2005. Antimicrobial susceptibility of *Escherichia coli* isolates from dairy farms using organic versus conventional production methods. J. Am. Vet. Med. Assoc. 226:589–594.
 49. Bandyopadhyay, S., J. Banerjee, D. Bhattacharyya, *et al.* 2018. Genomic identity of fluoroquinolone-resistant *bla* CTX-M-15 -type ESBL and pAmpC β -lactamase producing *Klebsiella pneumoniae* from buffalo milk, India. Microb. Drug Resist. 24:1345–1353.
 50. Kar, D., S. Bandyopadhyay, D. Bhattacharyya, *et al.* 2015. Molecular and phylogenetic characterization of multidrug resistant extended spectrum beta-lactamase producing *Escherichia coli* isolated from poultry and cattle in Odisha, India. Infect. Genet. Evol. 29:82–90.
 51. Duan, R.S., T.H.C. Sit, S.S.Y. Wong, *et al.* 2006. *Escherichia coli* producing CTX-M beta-lactamases in food animals in Hong Kong. Microb. Drug Resist. 12:145–148.
 52. Olowe, O.A., O. Adewumi, G. Odewale, O. Ojurongbe, and O.J. Adefioye. 2015. Phenotypic and molecular characterisation of extended-spectrum beta-lactamase producing *Escherichia coli* obtained from animal fecal samples in Ado Ekiti, Nigeria. J. Environ. Public Health 2015:497980.
 53. Wittum, T.E., D.F. Mollenkopf, J.B. Daniels, *et al.* 2010. CTX-M-type extended-spectrum β -lactamases present in *Escherichia coli* from the feces of cattle in Ohio, United States. Foodborne Pathog. Dis. 7:1575–1579.
 54. Pehlivanoglu, F., H. Turutoglu, D. Ozturk, and H. Yardimci. 2016. Molecular characterization of ESBL-producing *Escherichia coli* isolated from healthy cattle and sheep. Acta Vet. 66:520–533.
 55. Schmid, A., S. Hörmansdorfer, U. Messelhäuser, A. Käsbohrer, C. Sauter-Louis, and R. Mansfeld. 2013. Prevalence of extended-spectrum β -lactamase-producing *Escherichia coli* on Bavarian dairy and beef cattle farms. Appl. Environ. Microbiol. 79:3027–3032.
 56. Horton, R.A., L.P. Randall, E.L. Snary, *et al.* 2011. Fecal carriage and shedding density of CTX-M extended-spectrum β -lactamase-producing *Escherichia coli* in cattle, chickens, and pigs: implications for environmental contamination and food production. Appl. Environ. Microbiol. 77:3715–3719.
 57. Madec, J.-Y., C. Lazizzera, P. Châtre, *et al.* 2008. Prevalence of fecal carriage of acquired expanded-spectrum cephalosporin resistance in Enterobacteriaceae strains from cattle in France. J. Clin. Microbiol. 46:1566–1567.
 58. Hiki, M., M. Usui, A. Kojima, M. Ozawa, Y. Ishii, and T. Asai. 2013. Diversity of plasmid replicons encoding the *bla*(CMY-2) gene in broad-spectrum cephalosporin-resistant *Escherichia coli* from livestock animals in Japan. Foodborne Pathog. Dis. 10:243–249.
 59. Gautam, V., A. Thakur, M. Sharma, *et al.* 2019. Molecular characterization of extended-spectrum β -lactamases among clinical isolates of *Escherichia coli* & *Klebsiella pneumoniae*: a multi-centric study from tertiary care hospitals in India. Indian J. Med. Res. 149:208.
 60. Basu, S., S. Mukherjee, S. Roy, and A. Singh. 2011. CTX-M-9 group extended-spectrum β -lactamases in neonatal stool isolates: emergence in India. Indian J. Med. Microbiol. 29:305.
 61. Chetri, S., D. Bhowmik, D. Paul, *et al.* 2013. AcrAB-TolC efflux pump system plays a role in carbapenem non-susceptibility in *Escherichia coli*. BMC Microbiol. 19:210. <https://doi.org/10.1186/s12866-019-1589-1>
 62. Rahdar, H.A., E.S. Malekabad, A.-R. Dadashi, *et al.* 2019. Correlation between biofilm formation and carbapenem resistance among clinical isolates of *Klebsiella pneumoniae*. Ethiop. J. Health Sci. 29:745–750.
 63. Poiriel, L., C. Heritier, C. Spicq, and P. Nordmann. 2004. *In vivo* acquisition of high-level resistance to imipenem in *Escherichia coli*. J. Clin. Microbiol. 42:3831–3833.
 64. Webb, H.E., M. Bugarel, H.C. den Bakker, *et al.* 2016. Carbapenem-resistant bacteria recovered from faeces of dairy cattle in the high plains region of the USA. PLoS One 11:e0147363.
 65. He, T., Y. Wang, L. Sun, M. Pang, L. Zhang, and R. Wang. 2017. Occurrence and characterization of *bla*NDM-5-positive *Klebsiella pneumoniae* isolates from dairy cows in Jiangsu, China. J. Antimicrob. Chemother. 72:90–94.
 66. Mills, M.C., and J. Lee. 2019. The threat of carbapenem-resistant bacteria in the environment: evidence of widespread contamination of reservoirs at a global scale. Environ. Pollut. 255:113143.
 67. Bernasconi, O.J., V. Donà, J. Pires, *et al.* 2018. Deciphering the complete deletion of the *mgrB* locus in an unusual colistin-resistant *Klebsiella pneumoniae* isolate colonising the gut of a traveller returning from India. Int. J. Antimicrob. Agents 51:529–531.
 68. Azargun, R., M.R. Sadeghi, M.H. Soroush Barhaghi, *et al.* 2018. The prevalence of plasmid-mediated quinolone resistance and ESBL-production in Enterobacteriaceae isolated from urinary tract infections. Infect. Drug Resist. 11:1007–1014.
 69. Yamane, K., J.-I. Wachino, S. Suzuki, *et al.* 2007. New plasmid-mediated fluoroquinolone efflux pump, *qepA*, found in an *Escherichia coli* clinical isolate. Antimicrob. Agents Chemother. 51:3354–3360.
 70. Balakrishnan, S., P.X. Antony, H.K. Mukhopadhyay, *et al.* 2016. Genetic characterization of fluoroquinolone-resistant

- Escherichia coli* associated with bovine mastitis in India. Vet. World 9:705–709.
71. Hooper, D.C., and G.A. Jacoby. 2016. Topoisomerase inhibitors: fluoroquinolone mechanisms of action and resistance. Cold Spring Harb. Perspect. Med. 6:a025320.
 72. Paltansing, S., A.C. Tengeler, M.E.M. Kraakman, E.C.J. Claas, and A.T. Bernards. 2013. Exploring the contribution of efflux on the resistance to fluoroquinolones in clinical isolates of *Escherichia coli*. Microb. Drug Resist. 19:469–476.
 73. Das, A., C. Guha, U. Biswas, P.S. Jana, A. Chatterjee, and I. Samanta. 2017. Detection of emerging antibiotic resistance in bacteria isolated from subclinical mastitis in cattle in West Bengal. Vet. World 10:517–520.
 74. Granados-Chinchilla, F., and C. Rodríguez. 2017. Tetracyclines in food and feedingstuffs: from regulation to analytical methods, bacterial resistance, and environmental and health implications. J. Anal. Methods Chem. 2017. DOI: 10.1155/2017/1315497.
 75. Kyselková, M., J.J. Jirout, N. Vrchotová, *et al.* 2015. Spread of tetracycline resistance genes at a conventional dairy farm. Front. Microbiol. 6:536.
 76. Belaynehe, K.M., S.W. Shin, and H.S. Yoo. 2018. Inter-relationship between tetracycline resistance determinants, phylogenetic group affiliation and carriage of class 1 integrons in commensal *Escherichia coli* isolates from cattle farms. BMC Vet. Res. 14:340.
 77. Badi, S., P. Cremonesi, M.S. Abbassi, *et al.* 2018. Antibiotic resistance phenotypes and virulence-associated genes in *Escherichia coli* isolated from animals and animal food products in Tunisia. FEMS Microbiol. Lett. 365:10. DOI: 10.1093/femsle/fny088.
 78. Bandyopadhyay, S.S.S., I. Samanta, D. Bhattacharyya, *et al.* 2015. Co-infection of methicillin-resistant *Staphylococcus epidermidis*, methicillin-resistant *Staphylococcus aureus* and extended spectrum β -lactamase producing *Escherichia coli* in bovine mastitis—three cases reported from India. Vet. Q. 35:56–61.
 79. Hong, J.S., W. Song, H.M. Park, *et al.* 2019. First detection of New Delhi metallo- β -lactamase-5-producing *Escherichia coli* from companion animals in Korea. Microb. Drug Resist. 25:344–349.

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